

AURÉLIEN PONS

Mechanical Engineering – Composites, Structural Analysis, Design & Integration

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PROFESSIONAL SUMMARY

Final-year Mechanical Engineering student with three years of hands-on Formula Student experience across carbon fibre design, FEA, testing and team leadership. Helped Écurie Aix place 2nd at Formula Student Germany 2026 (400+ teams) with a stiffer, more reliable monocoque. Also builds Python, data and AI tools. Seeking internships and graduate roles, ready to relocate. French and Australian citizen based in Aachen, Germany.

KEY SKILLS

- **Composite Structures & FEA:** Altair HyperMesh / OptiStruct, laminate design, hand-calculation checks, correlation to test data
- **Composite Design & Manufacturing:** layup, autoclave and resin cure, void avoidance, ply cutting and nesting, mould tooling, FiberSim
- **CAD & Design Engineering:** Siemens NX (primary), SolidWorks (basic), packaging, engineering drawings, documentation
- **Testing & Fault Finding:** 3-point bending, push-through, cross-section failure analysis
- **Systems Integration & Leadership:** cross-team conflict resolution, led a 25-person team, requirements from rules and stakeholders
- **Programming, Data & AI:** Python, MATLAB, Power BI, Excel; builds AI-powered software

PROFESSIONAL EXPERIENCE

Chassis & Structural Design Engineer

09.2025 – Present

Écurie Aix, Formula Student Team, RWTH Aachen (2nd at Formula Student Germany 2026)

- Raised monocoque torsional stiffness by 75% (4,000 to 7,020 Nm/deg) for 35% more mass
- Ran FEA in Altair HyperMesh, checking every run with hand calculations first
- Correlated FEA to physical test data within 150 Nm/deg (~2% error)
- Found a resin-starvation risk by cross-sectioning a sample
- Rear wing FEA gave 11% less mass than the previous year and passed the rules check
- Resolved a packaging conflict between the structures and electrical sub-teams

Chassis Design Engineer (Composites)

09.2024 – 02.2025

University Racing Eindhoven

- Designed a carbon fibre monocoque from a clean sheet in Siemens NX (packaging, ergonomics, harness positions)
- Validated driver field of view on a purpose-built adjustable rig
- Designed mould tooling for demouldability (3° draft, alignment features)
- Ran 3-point bending and push-through tests on carbon fibre panels

Team Lead of Aerodynamics

06.2023 – 03.2024

Formula SAE, University of Technology Sydney

- Led a 25-person team designing aero components, including a rear diffuser, in Siemens NX with CFD

Business Intelligence & Automation Developer

2023 – Present

Oceania Distribution (part-time)

- Built a Python pipeline and Power BI dashboards; the analysis tool flagged over NZ\$2M in potential lost sales

EDUCATION

Bachelor of Engineering, Mechanical, Charles Darwin University, Australia

07.2025 – 06.2027 (Expected)

Bachelor of Engineering (Honours), Mechanical, University of Technology Sydney, Australia

02.2023 – 07.2025

Transferred to Charles Darwin University in 07.2025, credits recognised

Relevant coursework (UTS): Engineering Computations (High Distinction, 90), Materials and Manufacturing Engineering A (High Distinction), Mechanical Design Fundamentals Studio 1 (High Distinction)

Award: Faculty Academic Excellence Award, ENG423 (CDU), for an off-grid thermal comfort system for a remote community in Cape York

Languages: English (Native), French (Native), German (A2, learning) | **Citizenship:** French and Australian